

the latter (Xia, Li, & Li, 2023; Abramo, D'Angelo, & Felici, 2019). Finally, the use of citations requires reliance on bibliographic repertories, which do not index all publications and, in disciplines such as arts and humanities, their coverage is insufficient to provide a robust representation of research output (Aksnes & Sivertsen, 2019; Archambault et al., 2006; Moed, 2005).

The debate on which of the two approaches is more appropriate is still open and will likely remain so for a long time. There is no definitive answer: the choice between the two, either individually or in combination, will depend on the measurement goal, context, measurement scale, data availability, and resources and time constraints.

In the next section, we will analyse the non-scientific attributes of a scientific publication that, together with its intrinsic quality, determine its scholarly impact.

3. Non-scientific factors affecting publications impact

A comprehensive body of literature has emerged on the non-scientific factors influencing publication impact, as evidenced in the review by Tahamtan, Safipour Afshar, and Ahamdzadeh (2016). Xie et al. (2019) identified 66 factors possibly associated with impact.

These factors can be classified into two main categories: those external to the manuscript and those intrinsic to the manuscript. Among the former, notable elements include i) knowledge distribution channels such as the prestige level of the publishing journal (Mammola, Fontaneto, Martínez, & Chichorro, 2021; Bornmann & Leydesdorf, 2015; Stegehuis, Litvak, & Waltman, 2015) and the type of access (open access, or OA, vs non-open access, or non-OA) to the publication by a potential reader (Yu, Meng, Qin, Shen, & Hua, 2022; Langham-Putrow, Bakker, & Riegelman, 2021; Piwowar, et al., 2018; Wang, Liu, Mao, & Fang, 2015; Gargouri, et al., 2010; Antelman, 2004); and ii) communication initiatives in social media (i.e., blogs, Twitter, Facebook, pre-prints) undertaken by the authors to increase the visibility of the manuscript (Özkent, 2022).

Intrinsic factors within the manuscript can be further categorised into three groups based on the part of the manuscript they pertain to i) features of the byline, ii) features of the body of the manuscript, and iii) features of the reference list.

Regarding the features of the byline, factors associated with impact include i) the length of the author list (Thelwall et al., 2023b; Talaat & Gamel, 2022; Fox, Paine, & Sauterey, 2016; Abramo & D'Angelo, 2015; Didegah & Thelwall, 2013; Wuchty, Jones, & Uzzi, 2007); ii) the authors' academic influence and collaboration network (Hurley, Ogier, & Torvik, 2013; Mammola, Piano, Doretto, Caprio, & Chamberlain, 2022); iii) authors' gender or other personal features (Abramo, Aksnes, & D'Angelo, 2021; Andersen, Schneider, Jagsi, & Nielsen, 2019; Duch et al., 2012; Aksnes, Rorstad, Piro, & Sivertsen, 2011; Larivière, Vignola-Gagné, Villeneuve, Gelinas, & Gingras, 2011; Symonds, Gemmell, Braisher, Gorringer, & Elgar, 2006); iv) the number of institutions collaborating (Sanflippo, Hewitt, & Mackey, 2018; Narin & Whitlow, 1990); v) the number of countries involved (Glänzel & De Lange, 2002).

Moving to the features of the body of the text, factors linked to impact include i) document types (e.g., articles, reviews, proceedings papers, books, etc.), which are differently associated with impact and speed of impact (Wang, Song, & Barabási, 2013); ii) linguistic attributes of the manuscript (including title and abstract) such as readability (Ante, 2022; Heßler, Ziegler, 2022; Rossi & Brand, 2020; Stremersch, Camacho,

Vanneste, & Verniers, 2015; Didegah & Thelwall, 2013; Walters, 2006); iii) the manuscript's length (Xie, Gong, Cheng, & Ke, 2019; Elgendi, 2019; Fox, Paine, & Sauterey, 2016; Ball, 2008); iv) the degree of interdisciplinarity (Chen, Arsenault, & Larivière, 2015; Yegros-Yegros et al., 2015); v) popularity and interest of the subject (Peng & Zhu, 2012); vi) the discipline the manuscript falls under (Larivière & Gingras, 2010; Levitt & Thelwall, 2008); and vii) research fundings (Rigby, 2013).

Finally, concerning the reference list, factors associated with impact include i) the length of the reference list (Mammola, Fontaneto, Martínez, & Chichorro, 2021; Fox, Paine, & Sauterey, 2016); ii) the impact of the cited works (Sivadas & Johnson, 2015; Jiang, He, & Ni, 2013); iii) the incidence of more recent cited publications (Liu, Chen, Liu, Bu, & Gu, 2022; Mammola, Fontaneto, Martínez, & Chichorro, 2021); iv) the number of cited fields and their cognitive distance (Wang, Thijs, & Glänzel, 2015).

The non-scientific traits of a manuscript associated with its future impact are numerous, but not all of them are easily measurable in large-scale analyses.

4. Data and methods

4.1 Data

To better understand the methodology employed to address our research query, it is essential to delve into the VTR 2001-2003, which adopted a peer-review assessment. The primary objective of the VTR was to evaluate the research conducted by Italian universities and public research institutions during the specified timeframe. Each of the 102 institutions under scrutiny, encompassing 64,000 researchers, was tasked with submitting their research products during this timeframe. A restriction was imposed to ensure that the number of products did not exceed 50%⁴ of the full-time-equivalent (FTE) research staff of each evaluated university. Reviewers were entrusted with evaluating a comprehensive pool of 17,329 research products. Following the conclusion of the peer review process, each research product received a final judgment, expressed on a four-point rating scale: Excellent (E) = 1, denoting products that met the top 20% of international standards; Good (G) = 0.8 for those falling between 80% and 60%; Acceptable (A) = 0.6 for products scoring between 60% and 40%; and Limited (L) = 0.2 for products falling below the 40% threshold. Each product selected by universities and sent to the agency in charge of the evaluation was classified into a particular subject area (DAs, 14 in all). For the purposes of this paper, we have limited the analysis only to products (9,225 in all) classified in 7 of such areas and, more precisely: 1 - Mathematics and computer science; 2 - Physics; 3 - Chemistry; 4 - Earth sciences; 5 - Biology; 7 - Agricultural and veterinary sciences; 8 - Civil engineering; 9 - Industrial and information engineering. Notably, these cover all Sciences except Medicine.

These products are mainly journal articles (72%), but there are also books and book chapters (23%), patents (2%), and the remaining 3% of miscellaneous items.

In certain instances, co-authors from distinct institutions submitted identical products (in a very limited number of cases, in distinct DAs). Among the submitted products, 8,086 were scientific publications indexed in the Web of Science (WoS). Some of them were excluded since i) exhibited publication dates outside the 2001-2003 period; ii) were

⁴ To account for time devoted to teaching activities, 1 professor equals 0.5 FTE.

hosted in a source lacking *impact factor* (IF); or were assigned to different DA panels (as anticipated above), and received different evaluation scores. After eliminating all the aforementioned cases, the finalised dataset comprised 7,305 publications, whose breakdown by DAs is shown in Table 1. This corresponds to 11.8% of the WoS-indexed scientific production of all Italian academics within those DAs, ranging from a minimum of 7.2% in Industrial and information engineering to a maximum of 27.3% in Earth sciences.

Table 1. Number of Italian publications in the dataset by disciplinary area

Disciplinary area	Publications in the dataset (a)	Total Italian publications** (b)	a/b
1 - Mathematics and computer science	727	6,258	11.5%
2 - Physics	1,543	12,414	12.4%
3 - Chemistry	1,001	12,958	7.7%
4 - Earth sciences	581	2,126	27.3%
5 - Biology	1,504	14,545	10.3%
7 - Agricultural and veterinary sciences	667	4,053	16.3%
8 - Civil engineering	350	2,323	14.7%
9 - Industrial and information engineering	993	13,706	7.2%
Total*	7,305	61,592	11.8%

* The figures on the last line do not match the column total due to products falling into multiple DAs being counted more than once.

** Total 2001-2003 WoS publications authored by professors in each DA.

4.2 Methods

4.2.1 Variables

For the analysis, we focus on “journal articles” only, which are the vast majority of the research products in the dataset (6,889 articles out of 7,305). To assess the impact of the articles, we use WoS bibliometric data. Specifically, we calculate the normalised scholarly impact of publication i as its citations accrued up to 31/12/2022, normalised to the reference distribution, i.e. divided by the average number of citations (counted at the same date) received by all WoS publications classified in the same subject category (SC)^{5,6} and indexed in the same year of publication i . Having considered a fixed citation count date, the citation window used to assess the impact of publications varies from a minimum of 19 years to a maximum of 21. In all cases, these are extremely long, which ensures a reliable measurement of long-term scholarly impact. For ease of reading, in the following, we call the scholarly impact simply “impact”, and, being our target variable, we will denote it by Y . In fact, as will be clearer below, we will also consider the “short-term impact” of each publication, measured exactly as indicated above, but considering citations received on 31/12/2005, i.e., with the same time citation window available to the evaluators in the VTR.

We assume that the impact depends mainly on “quality”, denoted with Z , for the measurement of which we rely on the final judgment expressed by reviewers; hence,

⁵ The WoS classification scheme involves 255 SCs in all.

⁶ The SC of a publication corresponds to that of the journal where it is published. For publications in multidisciplinary journals (multiple SCs) the scaling factor is calculated as the average of the standardized values for each subject category.

quality is expressed through the four-point rating scale already described above (1 for “excellent products”; 0.8 for “good” ones; 0.6 for “acceptable” and 0.2 for “limited”).

On the other hand, as the literature suggests, we also assume that impact depends on several non-scientific factors ($X_1, \dots, X_p$). We group them into three sets of features.

Features of the byline:

- The number of authors of the publication.
- The average impact of their 2001-2003 publications (measured as the Y).
- A dummy variable for the presence of an English mother tongue author for modelling the possible “linguistic advantage”.
- The share of female co-authors.
- The number of institutions and countries in the affiliations list. The strong correlation between these two variables leads us to exclude both in favour of a single dummy (“foreign”) equal to 1 when the address list contains more than one country (0, otherwise).

Features related to the publication’s content and venue:

- The open-access character, expressed by a single dummy equal to 1 for Green, Hybrid, and Gold OA-tagged publications.
- The length of the publication expressed by the number of pages.
- Impact_factor of the hosting source, extracted from the Journal Citation Report 2004 edition and normalised to the reference distribution, i.e., divided by the average impact factor of all sources in the same subject category.
- The degree of interdisciplinarity of the publication measured by the share of cited papers in the bibliography falling in SCs other than the dominant one in the article reference list (Abramo, D’Angelo, & Zhang, 2018).

Features related to the publication’s bibliography:

- The length of the reference list (number of cited references).
- The share of references indexed in WoS signalling the extent of recourse to “qualified” literature.
- The share of self-citations in the reference list.
- The average age of cited publications.
- Their average normalised impact (measured as the outcome Y).

Some additional explanations deserve to be given on how the features related to the byline were measured. Specifically, each author of the publications in the dataset has been “disambiguated” using the algorithm proposed by Caron and van Eck (2014). This enables us to attribute their gender and nationality, survey their past scientific production, and measure the average impact of their 2001-2003 publications.

We dropped 220 articles with missing values on the authors’ average impact and 223 with outlying values in the number of authors, pages, or references. Hence, the final dataset for the analysis includes 6,446 articles.

Table 2 shows summary statistics of the variables of the dataset. For better readability, we omit all variables found to be not statistically significant at $p=0.05$ in terms of their effect on the response (Y), both here and in the following.